

Rodrigo Dornelles (MDS 2025)

LLM in Court:

Leveraging Open-Source Language Models to Enhance
Empirical Legal Research in Brazil



Context and Motivation: The Challenge



Legal documents are complex, requiring careful reading and manual tabulation

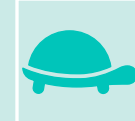


... expensive ...

Current methods demand significant time, resources, and attention to detail.



and (very!) slow...



Empirical legal research is scarce



Result: we do not test Law, just apply it

Example

SENTENÇA

processo Digital nº: 1519532-41.2024.8.26.0278
Classe – Assunto: Ação Penal - Procedimento Ordinário - Roubo
Documento de Origem: Comunicação de Prisão em Flagrante, Comunicação de Prisão em Flagrante, Comunicação de Prisão em Flagrante - 2234518/2024 - 72º D.P. VILA PENTEADO, 4211860 - 72º D.P. VILA PENTEADO, 2234518 - 72º D.P. VILA PENTEADO
Autor: Justiça Pública
Réu: VICTOR HUGO PEREIRA SAMPAIO

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MAYCON OLIVEIRA PIRES, qualificado nos autos, foi processado pela

suposta prática do delito descrito 180, *caput*, do Código Penal, porque, segundo denúncia: "consta, ainda, dos inclusos autos de inquérito policial que entre os dias 14 e 15 de agosto de 2024, nesta Capital, MAYCON OLIVEIRA PIRES, qualificado a fl. 15, adquiriu/recebeu, em proveito comum, o aparelho celular descrito conforme auto de exibição, apreensão e entrega de fls. 11, pertencente à vítima Felipe dos Santos Cruz, que sabia ser produto de crime" (folha 54).

O réu VICTOR HUGO foi preso em flagrante, tendo a Defesa requerido a concessão de liberdade provisória em seu favor 38/39 mas, em audiência de custódia, a prisão foi convertida em preventiva (fls. 44/46).

Auto de exibição, apreensão e entrega à folha 08.

Auto de exibição, apreensão e entrega às fls. 11/12.

Opportunity: AI as a Research Assistant



Automating the "mechanical" aspects using AI can make research faster and cheaper



This approach strengthens the scientific foundation of legal studies and informs policy



Particularly relevant in Brazil, one of the most litigious countries globally

Research Question

How can **open-source language models** be optimally deployed in the **Brazilian legal** context to accurately **extract information, recognize entities, and summarize** complex **legal documents**, considering factors such as cost, computational efficiency, and the specific challenges of legal language and document structure?

Preliminary Strategy

Literature review: SOTA of open source models

- Find models specialized to the Legal field
- Find models specialized in Portuguese

Select a specific problem

- Select data
- Manual build a golden dataset: from 50 to 100 cases
 - Validation / Test / FT

Prompt engineering

- Assess the quality of each model
- Compare in terms of energy, f1-score (balance and precision and recall), etc

Try to improve results

- Iteration over prompts
- Fine-tuning

Try to improve performance

- Pruning
- Distillation
- Other technics

Apply the best setting to the (whole) dataset

- Present a dataset ready to be analyzed by legal community
- Show first results in the poster

Specific Case-Study

- As a background to apply the methods
- Don't want to study in deep: it's not a research inside a research
- Brazil partial decriminalization on Drugs
- Supreme Court Decision
- June 2024
- Extract information about criminal sentences (I already have the data)



Questions?
Suggestions?
Loved it? Hated it?
LMK!!!

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